

Mohamed Kamel

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Education

Virginia Commonwealth University (VCU)

PH.D. IN COMPUTER SCIENCE

- GPA: 4.0/4.0

Richmond, VA, United States

2024 – Present

Nile University (NU)

M.Sc. IN INFORMATICS

- GPA: 3.97/4.0

Cairo, Egypt

2022 – 2024

Ahram Canadian University (ACU)

B.Sc. IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

- GPA: 3.9/4.0, Second of the class

Cairo, Egypt

2018 – 2022

Skills

Programming	Python, C++, Java, JavaScript Data Structures & Algorithms
ML Frameworks	PyTorch, TensorFlow, CUDA
ML & Generative AI	Deep Learning, Reinforcement Learning, Generative AI, Transformers, Diffusion Models, Rectified Flow, Flow Matching
Signal Processing	Signal Separation, Biomedical Signal Processing, Time Series Analysis, RF Signal Processing, Semantic Communication
Vision & Multimodal	Computer Vision, Medical Imaging, U-Net, ViT, Semantic Segmentation, Self-Supervised Learning, Multimodal Learning
Behavioral AI	Human Activity Recognition, Behavioral Sensing, Smart-Home Sensing, Digital Health
LLMs	Prompt Engineering, LoRA, NER, LLM Poisoning
Software & Backend	FastAPI, Docker, REST APIs, Microservices, MongoDB, Firebase
Tools & Cloud	Git, Linux, AWS, Azure
Languages	Arabic: Native English: C1 German: A1

Publications

BeatFlow-ECG: Rectified Flow for ECG Reconstruction from Indirect Wearable Signals

CONFERENCE PUBLICATION

ICLR

Under Review, 2027

TrustFlow-ECG: Reliability-Aware ECG Restoration with Calibrated Risk Estimation

CONFERENCE PUBLICATION

IEEE International Conference on
Acoustics, Speech, and Signal
Processing (ICASSP)

Under Review, 2026

FlowSep-RF: Rectified Flow Matching for Single-Channel Radio-Frequency Signal Separation

CONFERENCE PUBLICATION

IEEE International Conference on
Acoustics, Speech, and Signal
Processing (ICASSP)

Under Review, 2026

Communication-Aware Rectified Flow Matching for Robust Single-Channel RF Source Separation

JOURNAL PUBLICATION

IEEE Transactions on Wireless
Communications (TWC)

Under Review, 2026

SemTCP: Semantic-Aware Scheduling and Selective Recovery for Timely Task-Oriented Communication

CONFERENCE PUBLICATION

IEEE International Conference on
Communications (ICC)

Under Review, 2026

Joint User Activity Detection and Localization for Grant-Free Random Access in Cell-Free Massive MIMO Systems	<i>IEEE International Conference on Communications (ICC)</i>
CONFERENCE PUBLICATION	<i>Under Review, 2026</i>
ProSeD: Source-Side Semantic Decoys Against Unauthorized Data Harvesting	<i>Knowledge-Based Systems (KBS)</i>
JOURNAL PUBLICATION	<i>Under Review, 2026</i>
CondFlow-ECG: Conditional Rectified Flow for Morphology-Aware ECG Denoising	<i>International Conference on Machine Learning and Applications (ICMLA)</i>
CONFERENCE PUBLICATION	<i>Accepted, 2026</i>
Development of an AI-Enabled Remote Loneliness Assessment System for Older Adults Using Behavioral and Speech Data from IoT Sensors and Smart Speakers	<i>Assessment (SAGE), Special Issue on AI in Psychological Assessment</i>
JOURNAL PUBLICATION	[View Paper] 2026
Multimodal Activity Episode Inference from Smart-Home Signals for Behavioral Health Monitoring in Older Adults	<i>International Workshop on Machine Learning for Signal Processing (MLSP)</i>
CONFERENCE PUBLICATION	<i>Under Review, 2026</i>
URGE: Urgency-Aware Energy-Efficient UAV Mobility via Hybrid Deep Reinforcement Learning	<i>International Conference on Smart Mobility (SM)</i> [View Paper]
CONFERENCE PUBLICATION	2026
Attention-Driven Reinforcement Learning to Optimize Packet Retransmission in Semantic Communication	<i>IEEE GLOBECOM</i> [View Paper]
CONFERENCE PUBLICATION	2025
SkinFormer: Robust Vision Transformer for Automatic Skin Disease Identification	<i>JAC-ECC</i> [View Paper]
CONFERENCE PUBLICATION	2022

Academic Service

IET Communications journal

REVIEWER 2026

International Conference on Machine Learning and Applications (ICMLA)

REVIEWER 2026

Assessment journal

REVIEWER 2026

International Conference on Smart Mobility (SM)

REVIEWER 2026

Research Talks

From Wearable Signals to Clinical ECG: Toward Continuous Cardiac Monitoring

INVITED SPEAKER *Department of Computer Science Seminar, Virginia Commonwealth University*
March 20, 2026

Honors, Awards & Certifications

Badge issued by Virginia Commonwealth University (VCU)

LEADERSHIP FOUNDATIONS FOR GRADUATE STUDENTS

Credly

Oct 2025

Semi-Finalist, Dell Technologies Envision the Future

GRADUATION PROJECT COMPETITION (MIDDLE EAST, RUSSIA, AFRICA, TURKEY)

Online

2022

Certificate, Software Design and Architecture Specialization University of Alberta

COVERED: OBJECT-ORIENTED DESIGN, DESIGN PATTERNS, SOFTWARE ARCHITECTURE, SERVICE-ORIENTED ARCHITECTURE

Online

2021

First Place, ACU Problem Solving Competition

HACKEREARTH COLLEGE CHALLENGE

Leaderboard

2020

Academic Excellence Award

AWARDED MULTIPLE TIMES FOR TOP ACADEMIC PERFORMANCE

Ahram Canadian University

2019 – 2022

Experience

VCU + Bank of America – Special Olympics

SOFTWARE ENGINEER INTERN (PART-TIME)

United States

May 2025 – Aug 2025

- lead the backend development of a software application supporting Special Olympics.
- Involved in software design, API development, testing, and integration with frontend components.
- Tech Used: Python, FastAPI, MongoDB, Azure, Figma, React, TypeScript, HTML, CSS, JavaScript, Git

Virginia Commonwealth University (VCU)

GRADUATE ASSISTANT

United States

Jan. 2024 – Present

- **Research Assistant:** Contributing to the NIH-funded *HomePAL* project on AI-enabled health monitoring ([NIH R01AG072611](#), [Project Link](#)).
- Responsible for sensor setup, data acquisition scripts, signal analysis, and AI-based behavioral anomaly detection.
- Contributing to the SARE-funded food waste tracking and diversion project ([SARE FLW24-004](#), [Project Link](#)).
- **Teaching Assistant:** Held weekly office hours, graded assignments, and provided academic support for undergraduate CS courses.

Nile University (NU)

TEACHING ASSISTANT (FULL-TIME)

Egypt

Feb. 2023 – Jan. 2024

- Assisted in teaching AI, CS major, and programming courses.
- Provided lab support and one-on-one mentoring for students.

Ahram Canadian University (ACU)

TEACHING ASSISTANT (FULL-TIME)

Egypt

Oct. 2022 – Feb. 2023

- Taught AI-related and core programming subjects.
- Supported student learning and practical lab implementation.

LÀCRIMA Dairy Industrial Ltd

SOFTWARE DEVELOPER (CONTRACT)

Remote - Jordan

Jan. 2022 – Jul. 2023

- Developed an internal desktop system for electronic file archiving and communication ([Demo Video](#)).
- Designed customer-facing and employee mobile applications to improve service interaction. ([Android](#) | [iOS](#))
- Tech Used: Java, JavaFX, AWS, MySQL, Flutter, Firebase, Kotlin

Order GO

FULL STACK DEVELOPER (FULL-TIME)

Remote - United Kingdom

Apr. 2021 – Oct. 2021

- Built the company's public website and improved digital presence.
- Developed the ROS App for streamlining kitchen order management.
- Tech Used: Java, JavaFX, Kotlin, HTML, CSS, JavaScript, Bootstrap, PHP, Firebase, Git

Upwork

TOP-RATED FREELANCER

Freelance

Apr. 2019 – Oct. 2023

- Completed 65+ freelance software projects with a 98% Job Success Score.
- Delivered AI, web, mobile, and desktop solutions for international clients.

Volunteering Activities

Ahram Canadian University (ACU)

VOLUNTEER UNDERGRADUATE TEACHING ASSISTANT

Egypt

2018 – 2022

- Supported courses in Linear Algebra, Programming, Event-Driven Programming, and Data Structures; created educational videos, with one reaching nearly 60,000 views ([Video Link](#)).

EXCEPTION Student Committee

MEMBER AND ACADEMIC INSTRUCTOR

Egypt

2018 – 2022

- Conducted workshops in CS fundamentals, programming, Java (OOP), and Web development.

ACU CPC Community (ACM)

MENTOR

Egypt

2020 – 2022

- Mentored peers in programming competitions and problem solving.

ACU DSC Community (Google Student Developer)

MEMBER

Egypt

2020 – 2021

- Participated in technical workshops and peer-learning activities.

Selected Projects

NAVIGATING THE RETINAL MAZE: IR IMAGING FOR RETINAL EXPLORATION

2024

- Developed CrackRAVIR, a deep learning model for segmentation of retinal arteries and veins using IR imaging.
- Utilized Residual Attention UNET to improve segmentation accuracy and vascular width estimation.
- Improved glaucoma and AMD detection through detailed vasculature analysis.
- Technologies: Deep Learning, Residual Attention UNET, Semantic Segmentation, Python, IR Imaging.

GRAPH NEURAL NETWORKS FOR MOVIE RECOMMENDATION SYSTEMS

2024

- Developed a GraphSAGE-based recommendation system for movies using bipartite graphs of users/movies.
- Addressed cold start and scalability issues, compared with collaborative filtering and LightGCN.
- Technologies: Python, GraphSAGE, PyTorch Geometric, Collaborative Filtering, LightGCN.

EVIDENCE DETECTION IN CLOUD FORENSICS

2023

- Designed a system to detect evidence of cyber threats and anomalies in cloud environments.
- Utilized Gradient Boosting Machines, Sequential NN, and RNN to improve forensic accuracy and speed.
- Technologies: Python, GBM, RNN, Enhanced MLP.

AI DISEASE DIAGNOSTIC SYSTEM USING DEEP LEARNING AND IMAGE PROCESSING

2022

- Developed an application for early diagnosis of diseases using 50,000+ images from Fitzpatrick17k, SD198, and ISIC 2019 datasets.
- Achieved 85% accuracy using LAMB optimizer and advanced image augmentation techniques.
- Built with clean architecture and MVI framework.
- Technologies: Machine Learning, Deep Learning, Python, Kotlin, Vision Transformers, Neural Networks, MVI.
- [Demo Preview](#) | [Paper](#)

SELF-DRIVING CAR WITH OBJECT DETECTION (CNN, RL, YOLOv3)

2022

- Developed a self-driving car model capable of detecting traffic signals and adjusting speed dynamically.
- Technologies: Computer Vision, Carla Simulator, CNNs, Reinforcement Learning, YOLOv3, Python.

EXTENDING FASTPITCH WITH EMOTIONAL AWARENESS

2021

- Enhanced Fastpitch for emotional speech output using a modified Conformer architecture.
- Trained with emotional labels and perceptual loss for expressive speech synthesis.
- Technologies: Python, Conformer, Fastpitch, DeepVoice3, Mel Spectrogram.

EARLY-STAGE DIABETES RISK PREDICTION CLASSIFIER

2021

- Developed a binary classifier for diabetes risk prediction using tabular data.
- Technologies: Python, FastAI, Machine Learning. [GitHub Repo](#)

References

Available upon request.